Wheeler's Delayed-Choice Experiment and the Transactional Interpretation of Quantum Mechanics — TauTimp Formulation

Wheeler's Delayed-Choice Experiment (Truscott, Manning, Khakimov & Dall) and the Transactional Interpretation of Quantum Mechanics — TauTimp Formulation

Author: Viorel-Mihai Popescu **Framework:** TauTimp (Teoria Onduscularǎ) **Original working note:** *"Wheeler's delayed-choice gedanken experiment with a single atom"*, Beta version, 9 June 2025 **Regenerated:** 21 August 2026

Primary sources cited in the original note: - A. G. Manning, R. I. Khakimov, R. G. Dall, A. G. Truscott — *Wheeler's delayed-choice gedanken experiment with a single atom*, Nature Physics (local reference nphys3343.pdf). - J. G. Cramer — *The Transactional Interpretation of Quantum Mechanics and Quantum Nonlocality*, arXiv:1503.00039.

Abstract

Wave and particle behaviour are two mutually exclusive-seeming manifestations of light and matter, and the fact that the choice of measurement determines which of the two is observed remains one of the most counter-intuitive features of quantum mechanics. This note revisits Wheeler's delayed-choice gedanken experiment — realised with a single ultra-cold metastable helium atom in a Mach–Zehnder interferometer by Truscott and collaborators — through the lens of John Cramer's Transactional Interpretation of Quantum Mechanics (TIQM), in which a quantum process is mediated by a retarded *offer wave* and an advanced *confirmation wave* forming a single atemporal *transaction*. Within the TauTimp framework, the note assigns two explicit, distinct propagation speeds to the two wave sectors of the model: the **offer wave** (the physical, detectable photon or atom) propagates at the ordinary speed of light c, while the **confirmation wave** — identified here with the longitudinal *entanglement particle* that mediates nonlocal correlations between entangled quanta — propagates at $V = c \times 10^9$, the same superluminal interaction speed already adopted for the TauTimp gamma field in TauTimp_Gamma_Wave2.md. The distinction is essential: it keeps every directly observable signal strictly luminal, while confining the superluminal sector to a channel that carries no independently readable message.

1. Wave–particle duality and the role of measurement

Wave and particle behaviour are both natural manifestations of light and matter; the choice of measurement apparatus determines which of these seemingly incompatible behaviours is observed — a defining example of the counter-intuitive character of quantum mechanics.

Double-slit experiment. Photons interfere because they are temporarily ionised by tau-neutrino particles. The temporarily ionised photons can, in this picture, pass through the slits one at a time as a wave, and still build up the interference pattern.

Mach–Zehnder interferometer. Temporarily ionised photons travel, one at a time, along two paths — 50% one way, 50% the other. If the temporal separation between successive photons is shorter than one time quantum, they interfere with each other.

2. Temporal ionisation and neutrino energy thresholds

Nearly all particles carry spin (spin angular momentum — $\hbar/2$ for the electron) and are entangled through a particle-mediated **longitudinal wave** that propagates faster than light. In this formulation the propagation speed is fixed at

$$\boxed{V = c \times 10^9 = 2{,}998 \times 10^{17}\ \text{m/s}}$$

superseding the unspecified estimate "$V \gg c$" ($V \sim 10^{18}$ m/s) used in the original 2025 note; the two figures are of the same order of magnitude, and $V = c \times 10^9$ is retained here as the single, consistent value used throughout the TauTimp corpus.

The wavefunction collapses at measurement with photons in the optical window 400–700 nm, energy 2–4 eV — an energy that blocks the wave-to-particle transition because it exceeds the tau-neutrino threshold value of 0,925 eV. Measuring a single particle with an infrared sensor, at photon energy $h\nu$ below 0,925 eV, does not produce complete temporal ionisation, and generates no muon-neutrino unless the energy exceeds the threshold by at least 0,018 eV.

3. Wheeler's delayed-choice experiment

The phenomenon above is illustrated by Wheeler's famous delayed-choice gedanken experiment. It was recently realised, in a single-photon-equivalent configuration, using a single ultra-cold metastable helium atom in a Mach–Zehnder interferometer — an atomic analogue of Wheeler's original proposal (Truscott, Manning, Khakimov & Dall).

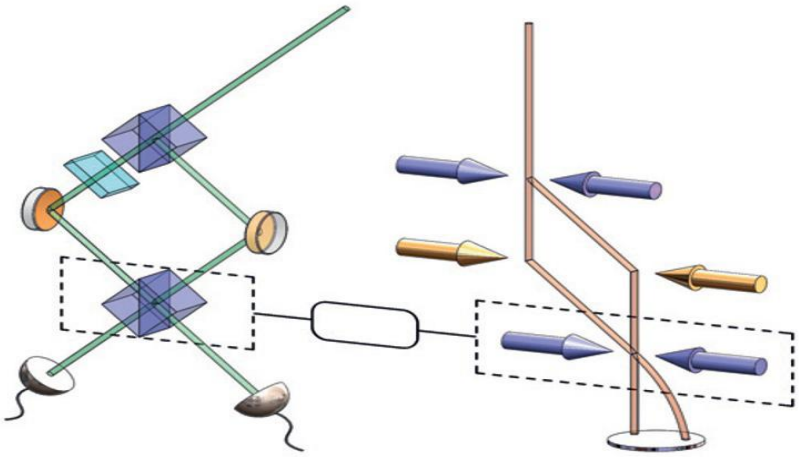


Fig. 1 — Schematic of Wheeler's delayed-choice experiment: the optical Mach–Zehnder interferometer (left) and its atomic analogue (right), after Manning, Khakimov, Dall & Truscott, Nature Physics (local reference nphys3343.pdf).

Truscott's experiment (2015). A single helium atom is sent into an interferometer, and the decision to insert or remove the second beamsplitter — i.e. whether interference is permitted — is made *after* the atom has already entered the interferometer.

4. The Transactional Interpretation of Quantum Mechanics (TIQM)

The essence of the Transactional Interpretation, as formulated by Cramer, is that a quantum process (e.g. the emission and absorption of a photon) occurs through an exchange of waves between an emitter and a receiver:

- The **retarded wave** (the ordinary wave) is sent forward in time by the source.
- The **advanced wave** is sent backward in time by the absorber.
- The two waves jointly form a **transaction**, a timeless agreement between cause and effect.

Following Cramer's original construction, with one explicit adjustment adopted here: the advanced wave, sent backward in time, is a **longitudinal wave** — the entanglement particle — and therefore propagates at $V = c \times 10^9$, not at an unspecified "much higher than c" rate.

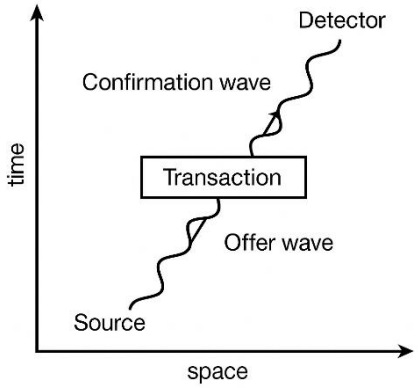


Fig. 2a — Space-time diagram of the transaction: the retarded offer wave (source to transaction) and the advanced confirmation wave (transaction to detector).

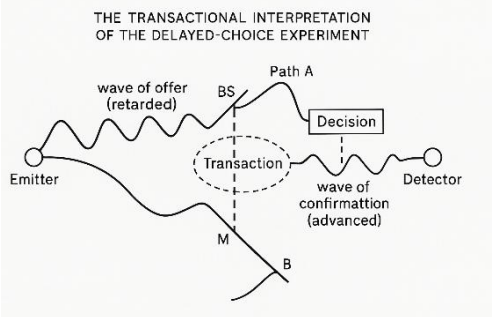


Fig. 2b — The Transactional Interpretation applied to the delayed-choice experiment: the offer wave (retarded) reaches the beamsplitter (BS); the confirmation wave (advanced) returns from the detector after the decision is registered, closing the transaction between emitter and detector via the mirror (M) / beamsplitter (B) paths.

Born rule:

$$P = |\psi(r,t)|^2 = \psi(r,t) \cdot \psi^*(r,t) \qquad \Rightarrow \qquad \text{Transaction} \sim \psi \cdot \psi^*$$

4.1 Transactional framework

- **Offer wave** ψ_O: propagates forward in time, from the emitter to all possible detectors, at speed c.
- **Confirmation wave** ψ_C: propagates backward in time, from a detector that might register the particle — sent only if absorption is possible — at speed $V = c \times 10^9$.
- **Transaction**: a standing wave formed between the offer and the confirmation wave, determining where the particle actually manifests.

4.2 Superluminal advanced-wave assumption

- The confirmation wave travels, at $V = c \times 10^9$, effectively instantaneously back to the emitter on laboratory timescales.
- This eliminates any measurable time lag in the emitter's "knowledge" of the future boundary conditions set at the detector.

5. Mathematical model

5.1 Offer wave from the emitter

The atom's wavefunction splits into two interferometer paths:

$$\psi_O = \frac{1}{\sqrt{2}}[\psi_A(x,t) + \psi_B(x,t)]$$

Each component acquires a phase set by the potential $V_A(x)$, resp. $V_B(x)$, along its path:

$$\psi_A(x,t) = e^{i(\varphi_A - \omega t)}, \qquad \varphi_A = \int_0^L k_A\,(x)\,dx, \qquad k_A(x) = \frac{\sqrt{2m(E - V_A(x))}}{\hbar}$$

$$\psi_B(x,t) = e^{i(\varphi_B - \omega t)}, \qquad \varphi_B = \int_0^L k_B\,(x)\,dx$$

5.2 Confirmation wave from the detector

Depending on whether the second beamsplitter is inserted, the detector sends back:

- a **coherent superposition**, enabling interference;
- a **single-path response**, disabling interference.

$$\psi_C = \frac{1}{\sqrt{2}}\left[e^{-i\varphi_A} + e^{-i\varphi_B}\right] \qquad \text{(interference enabled)}$$

$$\psi_C = e^{-i\varphi_A} \qquad \text{(interference disabled)}$$

This wave propagates back at $V = c \times 10^9$, letting the emitter effectively "know" the experimental configuration on a timescale far shorter than the light-crossing time of the apparatus.

5.3 The transaction

$$T = \psi_O \cdot \psi_C = \begin{cases} -\frac{1}{2}\left[1 + e^{i(\varphi_A - \varphi_B)}\right] & \text{(interference possible)} \\ -\frac{1}{\sqrt{2}} & \text{(no interference)} \end{cases}$$

Interference condition. The phase difference $\Delta\varphi = \varphi_A - \varphi_B$ determines constructive or destructive interference.

No interference. The transaction amplitude is then independent of the relative phase; the two paths are fully distinguishable.

6. Physical consequences

- The emitter receives effectively instantaneous feedback, via the advanced confirmation wave, regarding whether interference is permitted.

- Even though the choice is made after emission, the transaction forms only if there is consistent phase matching between the offer and confirmation waves.
- Retrocausality in the ordinary sense is avoided by treating the entire process as an atemporal negotiation among all boundary conditions, rather than a sequence of causally ordered events.

7. Key insight

The particle's "decision" to behave as a wave or as a particle is not made *at* the beamsplitter or the detector, but nonlocally and atemporally, through the matching of the offer and confirmation waves across the entire spacetime configuration of the experiment — including settings fixed only in the future.

Superluminal transactional laws. All particles communicate bidirectionally in time through an otherwise undetectable channel, at $V = c \times 10^9$ for the confirmation/entanglement sector — never for the directly observable photon, which remains bound to c.

Atemporal partial determinism. Past and future are co-determined; events "settle" according to overall coherence across the full boundary-value problem, not by strict forward propagation alone.

Active non-locality. Distant events can influence a local transaction through this invisible channel.

Locally reversible causality. Effects may appear to precede their apparent cause, but only insofar as this is consistent within a single, completed transaction.

8. Summary of numerical values

Quantity	Symbol	Value
Speed of light (offer/retarded wave, the observable photon)	c	$2{,}998 \times 10^8$ m/s
Entanglement-particle speed (confirmation/advanced wave)	V	$c \times 10^9 = 2{,}998 \times 10^{17}$ m/s
Tau-neutrino energy threshold	—	$0{,}925$ eV
Additional threshold for muon-neutrino generation	—	$+0{,}018$ eV
Optical collapse window	—	400–700 nm (2–4 eV)

Note on the change from the original note

The 2025 working note used an order-of-magnitude estimate, "$V \gg c$ ($V \sim 10^{18}$ m/s)", for the entanglement-particle speed. This regenerated version fixes that value at the single TauTimp-canonical figure $V = c \times 10^9 \approx 2{,}998 \times 10^{17}$ m/s (already adopted in TauTimp_Gamma_Wave2.md for the gamma field), so that every document in this corpus refers to the same numerical constant. The speed of light c is left untouched throughout for every directly observable wave (the offer wave, i.e. the physical photon or atom); only the advanced/confirmation-wave sector — which carries no independently readable message, per the no-signalling requirement discussed in Rezumat_Pettini_Entanglement_RO.md — is assigned the superluminal value.

References

1. Manning, A. G., Khakimov, R. I., Dall, R. G., Truscott, A. G. — *Wheeler's delayed-choice gedanken experiment with a single atom*, Nature Physics (local reference nphys3343.pdf).
2. Cramer, J. G. — *The Transactional Interpretation of Quantum Mechanics and Quantum Nonlocality*, arXiv:1503.00039.
3. Popescu, V.-M. — *TauTimp — Gamma-Ray Longitudinal Wave: Physical Foundations and Full Field-Theoretic Derivation*, TauTimp_Gamma_Wave2.md (source of $V = c \times 10^9$).

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